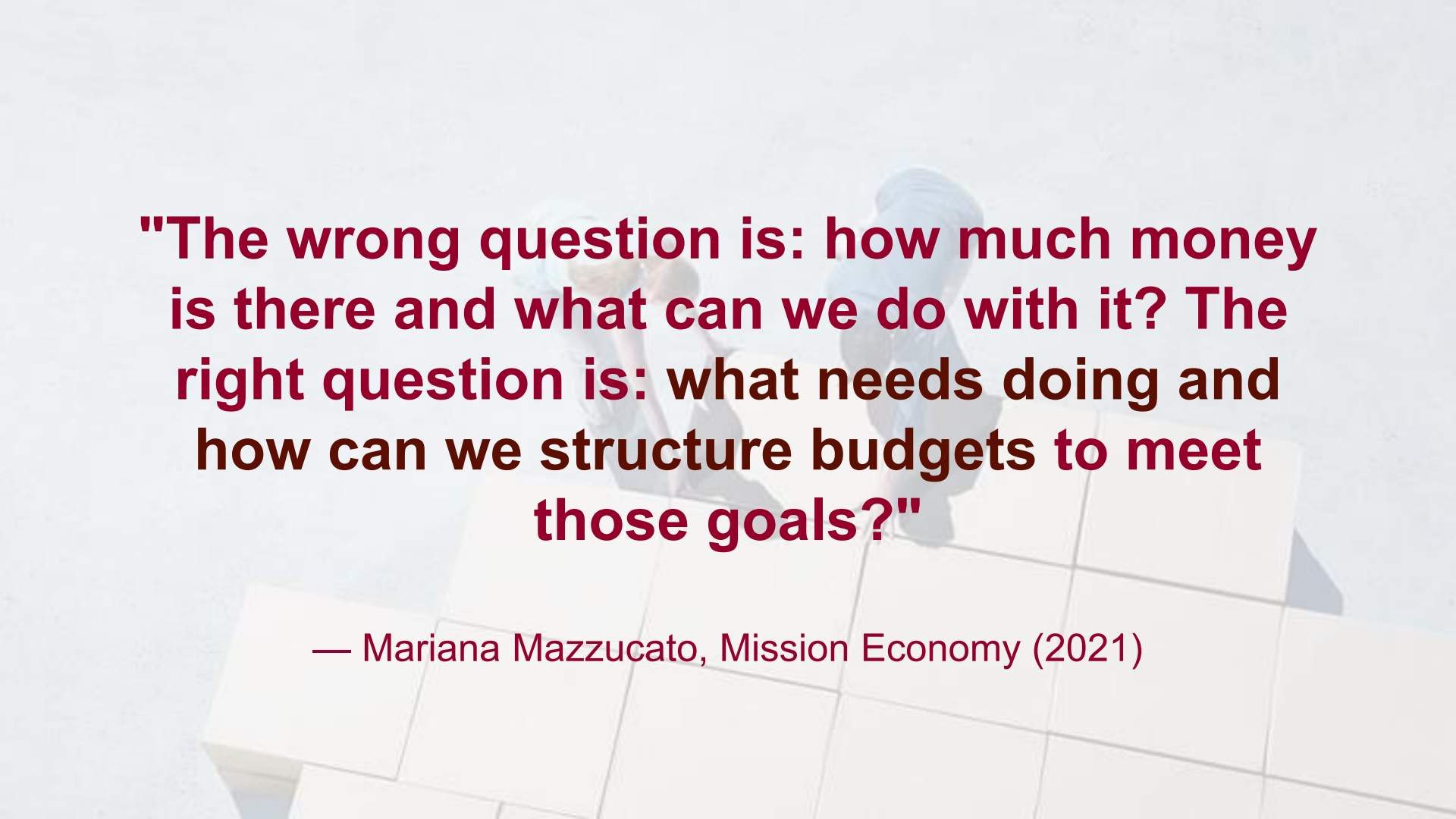


Measuring and optimizing the industry-level impact of government spending



Kamau Njendu, Chris Howard

The background of the slide features a high-angle, slightly blurred photograph of two individuals, a man and a woman, working together on a large-scale project. They are positioned on a floor made of large, light-colored square tiles. The man, wearing a blue shirt, is leaning over and placing or adjusting a block. The woman, wearing a light-colored top, is also leaning over, focused on the task. The overall scene suggests a collaborative effort in a professional or academic setting, possibly related to architecture, engineering, or a large-scale data visualization project.

"The wrong question is: how much money is there and what can we do with it? The right question is: what needs doing and how can we structure budgets to meet those goals?"

— Mariana Mazzucato, Mission Economy (2021)



Motivation & Literature Review

- Government spending is large but not optimized for impact
- Prior work evaluates spending, but does not prescribe allocation decisions

Data Collection & Cleaning

- BEA industry-level economic data (output, employment, value-added)
- USA Spending contract data (NAICS-level government investment)
- Industry-level panel-construction (2009-2025)

Modeling

- Panel regression with industry and year-fixed effects
- Estimation of government spend multipliers
- Mixed effects model for industry-specific impact heterogeneity

Optimization

- Optimize allocation to maximize economic impact
- Compare current vs optimal spending allocation

Key Takeaways & Next Steps

- Government spending impact varies significantly across industries
- Current allocation is not aligned with highest return sectors
- Reallocation can increase economic impact without increasing the budget



Motivation

The U.S. government spends trillions annually, yet the economic impact of these investments is poorly understood:

- Policy-makers are prioritizing the reduction of inefficient or low-impact spending
- National debt continues to rise despite sustained government investment
- The economic impact of investment is difficult to measure at the industry-level

This raises a key question: how should government spending be allocated to maximize economic impact?



Literature Review

The research findings below heavily influenced our decision to analyze government spend:

- **Nakamura & Steinsson (2014):** Using regional variation in U.S. military spending, the authors estimate a fiscal multiplier of ~ 1.5 , showing that government demand shocks can have large effects on output
- **Chodorow-Reich (2019):** Reviewing cross-sectional fiscal multiplier studies alongside new ARRA analysis, the paper estimates a national spending multiplier of 1.7 or above, affirming the significant economic impact of government expenditure
- **Mazzucato (2021):** Argues that governments create the most economic value not by merely fixing market failures, but by acting as bold, mission-driven investors that actively shape and co-create markets alongside the private sector

However... industry-specific multipliers have not been largely studied



Project goal



We optimize government spending by allocating funds to industries with the highest estimated economic return per dollar spent



Data Collection

USA Spend:

- Contains all public federal spend from 2001-2026 including contracts, loans, grants, direct spend, etc.,
- We collected 10,000 contracts per fiscal year over the period 2009-2025, and sorted these contracts by NAICS codes, also keeping track of recipients name, transaction amounts, and awarding agencies

BEA:

- Pulled BEA industry-level economic data ranging from 1997-2025 sorted by NAICS code and fiscal year
- We recorded NAICS codes, fiscal years, value added, gross output, and employment to capture industry performance across the period

Panel Data:

- The BEA and USASpend data were aggregated using fiscal year and NAICS code to create an industry-level panel (2009-2025)
- Created the key variable gov spend intensity which we define as gov spending divided by gross output to make industries comparable
- Final data structure:

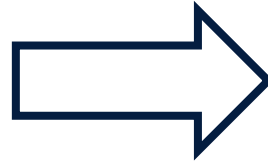
*NAICS x Fiscal_Year x Gov_Spend x Value_Added
x Employment x Intensity*



Data Cleaning - Handling Missing Industry-Year Data

Identified missing spend observations across industries and years

fiscal_year	2009	2010	2011	2012	2013
naics_bea					
11	1.910392e+07	8.729887e+06	NaN	1.149957e+07	7.400000e+06
211	4.847495e+07	1.587075e+08	8.810596e+07	NaN	9.900000e+06
212	NaN	6.495000e+07	NaN	NaN	5.491495e+08
213	NaN	NaN	NaN	NaN	6.273359e+06
22	1.229424e+09	7.757765e+08	9.923235e+08	7.206121e+08	6.133156e+08

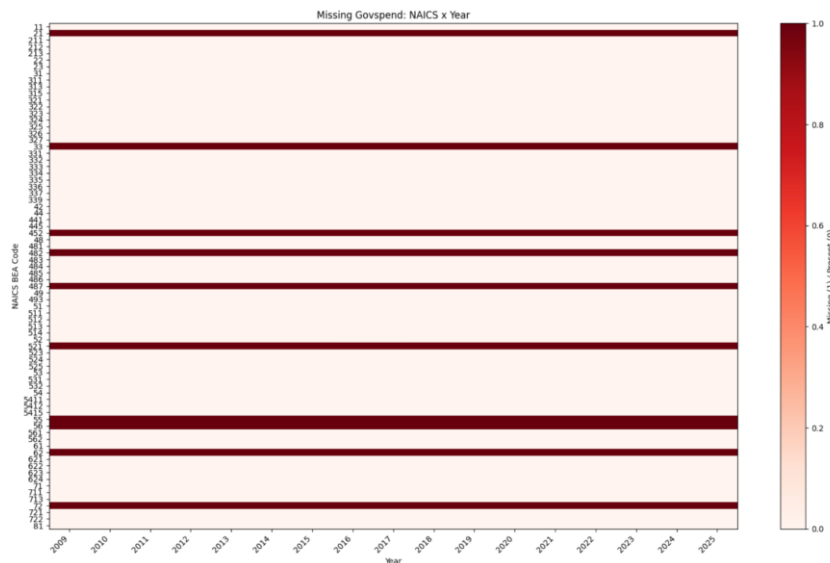


Visualized missing observations by industry



Data Cleaning - Imputation

Applied within-industry linear interpolation
to impute missing spending values



Modeling...

1. TWFE Regression
2. Hierarchical Mixed Effects

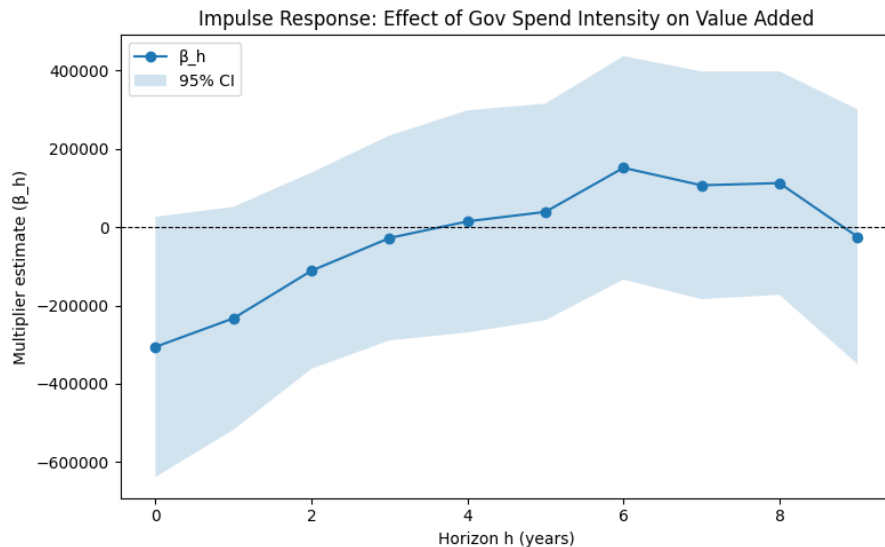


Vocabulary to know

- **Value Added** - Measures industry contribution to GDP
 - Total industry outputs minus intermediate inputs
- **Value Added Growth** - Year-over-year change in value added
- **Employment** - Captures labor market impact of industrial activity
 - Total number of workers in a given industry
- **Government Spending Intensity** - Measures how large government spending is relative to industry size
 - $\text{Intensity} = \text{Government spending} \div \text{industry gross output}$
- **Gross output** - Total value of goods and services produced by an industry
- **Multipliers** - Measures return on government spending
 - Change in future value added associated with an increase in government spending



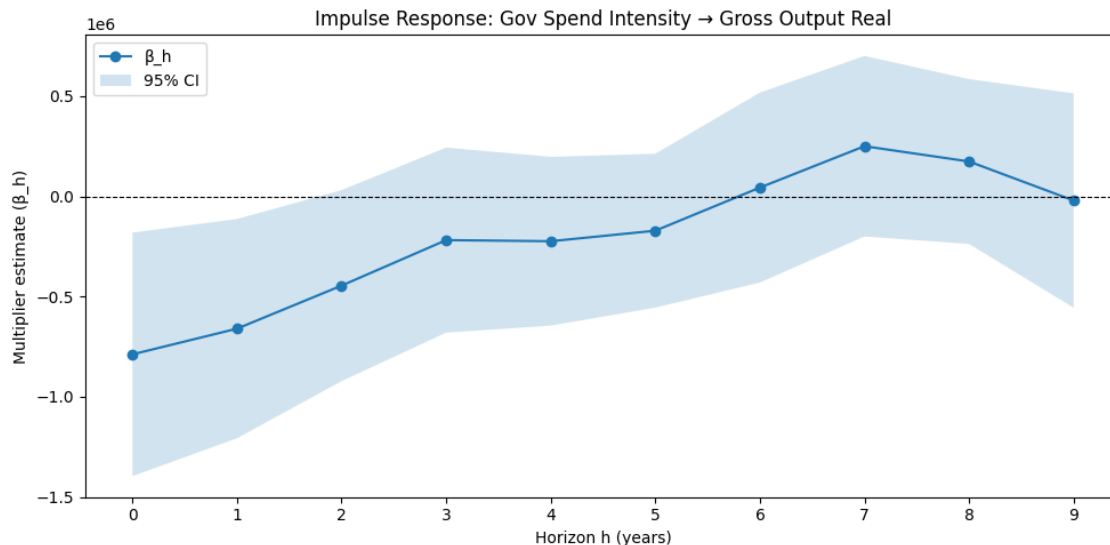
Dynamic Effects of Government Spending: TWFE Regression (Value Added)



- **Ran a two-way fixed-effect regression**
 - The model estimates the impact of government spending on value added over horizon h , controlling for industry and year-specific factors
- **Effects are delayed and peak 5-6 years after spending**
 - Government spending has a delayed, yet positive effect on industries



Dynamic Effects of Government Spending: TWFE Regression (Gross Output)

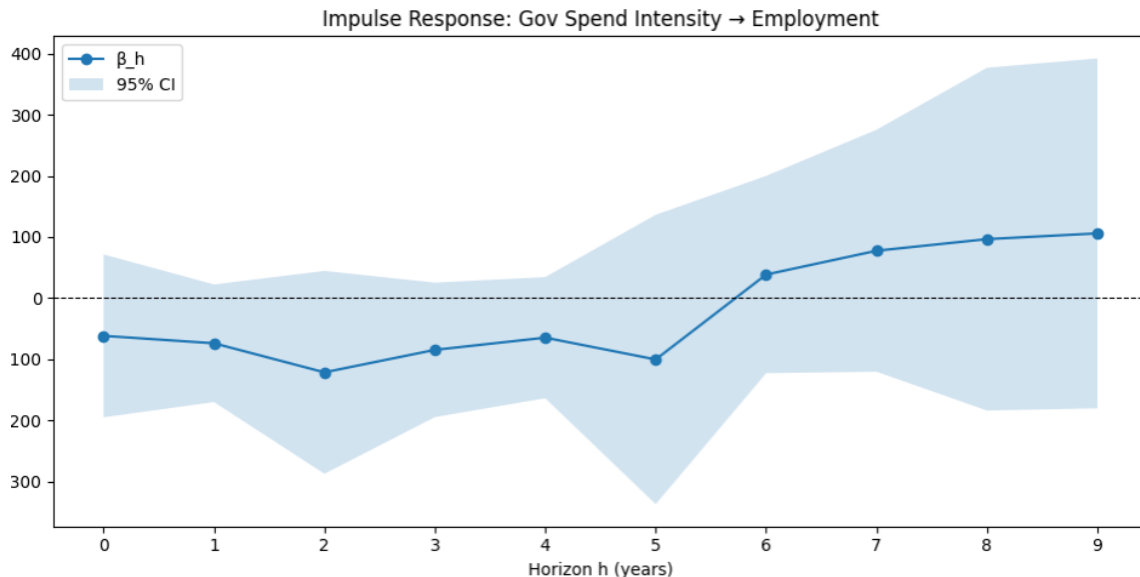


- **Interpretation**

- Each point shows effect h years after spending
- Uses with industry variation over time
- Effects peak 6-7 years after spending



Dynamic Effects of Government Spending: TWFE Regression (Employment)



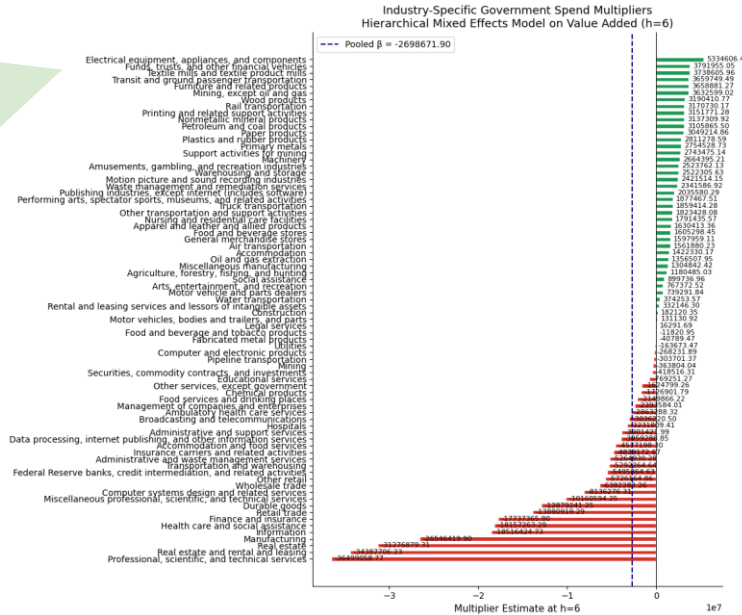
- **Interpretation**

- Similar to value-added and gross output we see that the impact of spend on employment is delayed
- The larger lag suggests that firms first adjust production and then modify employment



Dynamic Effects of Government Spending: Hierarchical Mixed Effects Model on Value Added

Electrical
equipment,
Financial
vehicles,
Ground
Passenger
Transportation



Professional
services, Real
estate,
Manufacturing

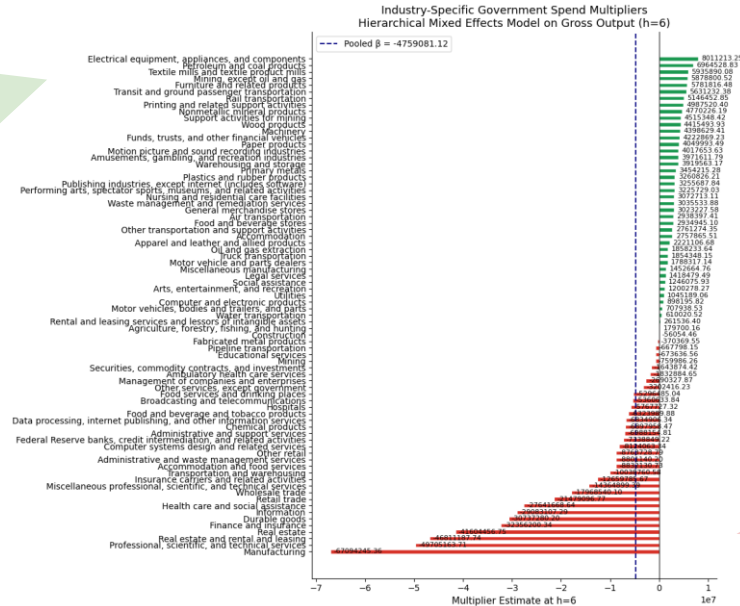
Model

- Hierarchical mixed effects model with industry-specific slopes
- Multipliers vary significantly across industries
- Multipliers represent the estimated change in future value added per unit increase in government spending intensity



Dynamic Effects of Government Spending: Hierarchical Mixed Effects Model on Gross Output

Electrical equipment,
Petroleum
products,
Ground Textile
Mills



Manufacturing,
Professional
services, Real
estate

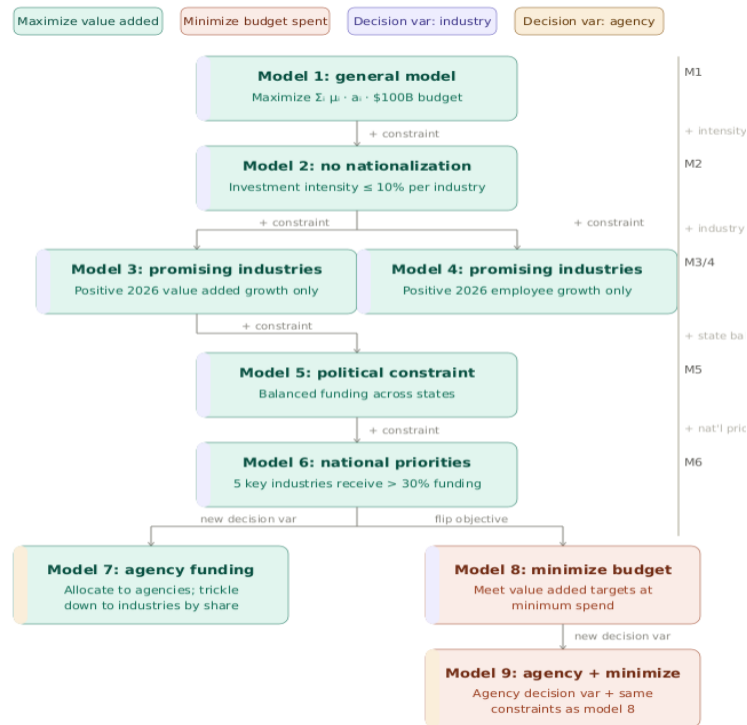
• Industry Effects

- Significant variation across industries
- Varied output enables targeted allocation of government spending



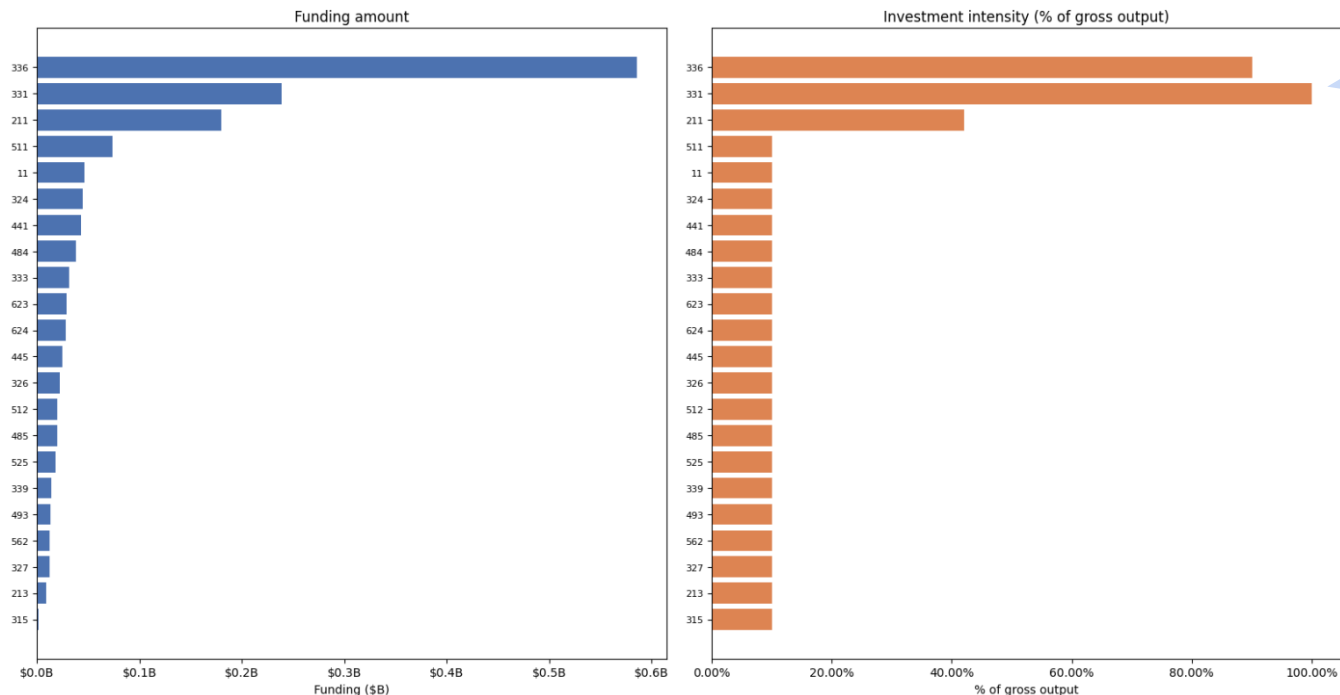
Model hierarchy: From baseline optimization to policy-constrained allocation

- We start with a baseline model to maximize economic impact
- We progressively introduce constraints to make the model more realistic
 - Political / geographical balance
 - Investment intensity
 - Limits on concentration
- We explore different decision frameworks
 - Allocation by agency / industry
 - minimizing budget for a given target



Final Model

Model 8: Min budget for 95% of max VA (\$1.5B)



- Transportation Equipment Manufacturing
- Primary Metal Manufacturing
- Oil and Gas Extraction

- A large percentage of funding is allocated to a few sectors
- Once we fund top industries, additional funding in lower-return sectors contributes marginally to the total value added



Full Model Optimization Formulation (M8)

1. Sets

I = set of industries, A = set of agencies, S = set of states

2. Parameters

μ_i = estimated effect (multiplier) for industry i

B = total available budget

G_i = gross output of industry i

c = maximum allowed investment intensity

$$1_i^{VA} = \begin{cases} 1 & \text{if industry } i \text{ has positive value-added growth} \\ 0 & \text{otherwise} \end{cases}$$

$$1_i^{EMP} = \begin{cases} 1 & \text{if industry } i \text{ has positive employment growth} \\ 0 & \text{otherwise} \end{cases}$$

$P \subseteq I$ = set of priority industries

$s(i)$ = state associated with industry i

$w_{a,i}$ = share of agency a funding going to industry i

T = target level of value added (for minimization models)



Full Model Optimization Formulation (M8)

3. Decision Variables

Industry-level allocation (Models 1–6, 8)

$n_i \geq 0$ investment intensity in industry i

$$a_i = n_i G_i$$

Agency-level allocation (Models 7, 9)

$b_a \geq 0$ spending allocated to agency a

$$n_i = \frac{1}{G_i} \sum_{a \in A} w_{a,i} b_a$$



Full Model Optimization Formulation (M8)

4. Objective Functions

Models 1–7: Maximize economic impact

$$\max \sum_{i \in I} \mu_i n_i$$

Models 8–9: Minimize total budget

$$\min \sum_{i \in I} n_i G_i \quad \text{or} \quad \min \sum_{a \in A} b_a$$



Full Model Optimization Formulation (M8)

5. Constraints by Model

Model 1: Baseline budget constraint

$$\sum_{i \in I} n_i G_i \leq B$$

Model 2: Investment intensity cap

$$0 \leq n_i \leq c \quad \forall i \in I$$

Model 3: Positive value-added growth industries only

$$n_i = 0 \quad \text{if } 1_i^{VA} = 0$$

Model 4: Positive employment growth industries only

$$n_i = 0 \quad \text{if } 1_i^{EMP} = 0$$



Full Model Optimization Formulation (M8)

Model 5: State balance constraint

$$\sum_{i:s(i)=s} n_i G_i \approx \frac{B}{|S|} \quad \forall s \in S$$

Model 6: National priority industries

$$\sum_{i \in P} n_i G_i \geq 0.3B$$

Model 7: Agency-level allocation

$$n_i = \frac{1}{G_i} \sum_{a \in A} w_{a,i} b_a$$
$$\sum_{a \in A} b_a \leq B$$



Full Model Optimization Formulation (M8)

Model 8: Minimize budget subject to impact target

$$\begin{aligned} \min \quad & \sum_{i \in I} n_i G_i \\ \text{s.t.} \quad & \sum_{i \in I} \mu_i n_i \geq T \end{aligned}$$

Model 9: Agency-level + minimize budget

$$\begin{aligned} \min \quad & \sum_{a \in A} b_a \\ \text{s.t.} \quad & \sum_{i \in I} \mu_i \left(\frac{1}{G_i} \sum_{a \in A} w_{a,i} b_a \right) \geq T \end{aligned}$$



Key Takeaways



Optimal spending is highly concentrated



Diminishing returns are significant



Multipliers drive allocation



Investment intensity reveals saturation points



Thank you!

